



HOMEWORK TASK 2

FINANCIAL TECHNOLOGY MODELS

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READING ASSIGNMENT

The given paper, “*Statistical Modeling: The two cultures*” (2001), presents a fundamental dichotomy in data analysis, challenging the statistical community’s reliance on traditional methods in the face of increasingly complex data. The division centers on the core assumption made about the function that relates input variables to the response.

TRADITIONAL STATISTICS

This culture is defined by its foundational belief in a stochastic data model. Analysts assume that the data generated by a known, structured model where the effects are additive and the parameters are directly interpretable.

- **Foundational assumption** → Presumes that data arise from a known stochastic model with a structured and often additive form. The analyst specifies this model a priori, and its parameters are intended to possess clear interpretability.

Seeks to understand the functional relationships between variables by estimating model parameters and drawing inferential conclusions about the presumed data-generating mechanism.

- **Modeling Paradigm** → relies predominantly on parametric models chosen for their simplicity, interpretability, and theoretical grounding. The selected model is treated as a meaningful representation of the underlying natural process.
- **Assessment of validity** → evaluates model adequacy through internal criteria such as goodness-of-fit tests, residual diagnostics, and other procedures designed to verify whether assumptions hold.
- **Breiman’s critique** → emphasizes the vulnerability of this culture to model misspecification. When the chosen model fails to capture the true complexity of the data-generating mechanism, subsequent inference and interpretation become unreliable, potentially leading to misleading scientific conclusions.

MODERN MACHINE LEARNING

This culture operates under the assumption that the true data generating process is an unknown and highly complex black box. It shifts focus from interpreting the model’s internal structure to maximizing its predictive utility.

- **Foundational assumption** → operates under the premise that the true data generating mechanism is unknown and highly complex. The focus shifts from specifying a model structure to allowing algorithms to learn patterns directly from the data.

Prioritizes prediction, aiming to construct a function that minimizes error on new, unseen data. Interpretability is secondary to predictive performance.

- **Modeling paradigm** → employs flexible, non-parametric or weakly parametric algorithms such as random forests, neural networks, and boosting methods. These models adapt to high dimensional, nonlinear relationships without imposing rigid functional assumptions.
- **Assessment of validity** → measures success externally through predictive accuracy, typically evaluated via generalization error, test-set performance or cross validation. The superior model is the one that predicts most effectively.
- **Justification for complexity** → argues that complex modeling strategies are often required to capture the nonlinearities and interactions present in real world data. Although such models may lack immediate transparency, Breiman contends that their empirical performance provides more reliable scientific insight than adherence to overly simplistic theoretical models.

The distinction made by this paper emphasizes the fundamental difference between traditional statistical modelling and modern machine learning. Traditional statistical methods are typically based on parametric models that are easily understood and easily interpreted by humans; these models are sometimes too simple to capture the actual richness of a given phenomenon or process. By contrast, machine learning is generally focused on high levels of accuracy from statistical models—and therefore, high levels of predictive ability—from flexible computer programs. As it was shown in this article, due to the extreme complexity of many modern datasets, there continues to be significant opportunities for empirical performance to become the primary focus of model selection, along with the continued growth in popularity of machine learning algorithms wherever appropriate.

CALCULATION QUESTIONS

2. Show that in OLS the matrix of second derivatives $H = 2X^T X$ is positive definite.

Let $H = 2X^T X$, H is the Hessian of the residual sum of squares RSS, so positive definiteness implies strict convexity and a unique minimizer. We also have to assume that X has full column rank otherwise H is only positive semidefinite.

For any vector $v \in \mathbb{R}^p$,

$$v^T H v = 2v^T X^T X v = 2(Xv)^T (Xv) = 2 \|Xv\|^2$$

Therefore $v^T H v \geq 0$ for all v , so H is positive semidefinite. Moreover, if X has full column rank (rank p), then $Xv = 0$ implies $v = 0$, so for every nonzero v we have $\|Xv\|^2 > 0$. Thus $v^T H v > 0$ for all $v \neq 0$ and H is positive definite.

If X is not full rank, H is only positive semidefinite.

3. In weighted least squares we minimize the objective function:

Let $J(b)$ be:

$$\sum_{i=1}^n w_i (y_i - x_i^T b)^2$$

For a set of positive weights $w_1, \dots, w_n$. This enables us to introduce different "importance" of observations during the fitting process.

- Show that the objective function can be expressed as $\|D(Xb - y)\|^2$ for some matrix D . Find the expression for matrix D .
- Show that if the columns of X are linearly independent, then the columns of DX are also linearly independent.
- Derive the matrix solution of parameter vector $\tilde{b}$ when weighted least squares objective function is minimized. Hint: use the similarity of $\|DX\tilde{b} - Dy\|^2$ with the ordinary least squares objective function.

(1) Write the objective as a norm $\|D(xb-y)\|^2$ and find D

Let $W = \text{diag}(w_1, \dots, w_n)$. Define $D = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})$. Then:

$\|D(Xb-y)\|^2 = (Xb-y)^T D^T D (Xb-y) = (Xb-y)^T W (Xb-y)$ But the scalar $(Xb-y)^T W (Xb-y)$ expands to:

$$\sum_{i=1}^n w_i (x_i^T b - y_i)^2 = \sum_{i=1}^n w_i (y_i - x_i^T b)^2 = J(b)$$

So the required matrix is $D = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})$

2. If columns of X are linearly independent then columns of DX are linearly independent

Assume the columns of $X \in \mathbb{R}^{n \times p}$ are linearly independent so $Xv = 0 \Rightarrow v = 0$. Because each $w_i > 0$ the diagonal matrix D has strictly positive diagonal entries and is therefore invertible.

If $DXv = 0$ for some v , multiply D^{-1} to get $Xv = 0$. By independence of X 's columns, $v = 0$. Hence the columns of DX are linearly independent.

(3) Matrix solution $\tilde{b}$ minimizing the weighted objective

Minimize $J(b) = (Xb-y)^T W (Xb-y)$ Differentiate with respect to b and set gradient zero:

$$\nabla_b J(b) = 2X^T W (Xb-y) = 0$$

So, the normal equations for weighted least squares are

$X^T W X \tilde{b} = X^T W y$ if $X^T W X$ is invertible (this holds when X has full column rank and all $w_i > 0$ then

$$\tilde{b} = (X^T W X)^{-1} X^T W y$$

→ using D from above, this is the ordinary least squares solution on the transformed data:

$$\tilde{b} = ((DX)^T (DX))^{-1} (DX)^T (Dy)$$

5. Show that the derivative of the logistic function $f(u) = 1/(1+e^{-u})$, is $f'(u) = f(u)(1 - f(u))$.

To show that

$$f(u) = \frac{1}{1+e^{-u}} \implies f'(u) = f(u)(1-f(u))$$

we must differentiate directly:

(1) Differentiate $f(u)$

$$f(u) = (1+e^{-u})^{-1}$$

$$f'(u) = -(1+e^{-u})^{-2} \cdot (-e^{-u}) =$$

$$= \frac{e^{-u}}{(1+e^{-u})^2}$$

(2) Express the derivative in terms of $f(u)$

$$f(u) = \frac{1}{1+e^{-u}} \quad 1-f(u) = \frac{e^{-u}}{1+e^{-u}}$$

$$f(u)(1-f(u)) = \frac{1}{1+e^{-u}} \cdot \frac{e^{-u}}{1+e^{-u}} = \frac{e^{-u}}{(1+e^{-u})^2}$$

- Since both derivatives are the same, we have proved that the statement is true.

6. Consider the decision tree on lecture slides do the following:

Change in impurity

Weighted Gini of children:

$$\frac{54}{100} * 0.168 + \frac{46}{100} * 0.043 = 0.1105$$

Gini reduction due to the split:

$$\Delta Gini = Gini(parent) - \text{weighted Gini(children)} = 0.5 - 0.1105 = 0.3895$$

So the split reduces Gini by 0.3895.

Change in impurity using entropy

Parent entropy (proportions 0, 0.5, 0.5):

$$H_{parent} = 1.0 \text{ bit}$$

Children entropies:

- Left child ([0,49,5]): $H_l = 0.4450649$
- Right child ([0, 1, 45]): $H_r = 0.1510970$

Weighted child entropy:

$$\frac{54}{100}H_l + \frac{46}{100}H_r \approx 0.3098396$$

Entropy reduction:

$$\Delta_{ent} = H_{parent} - \text{weighted children} \approx 1.0 - 0.3098396 - 0.6901504 \text{ bits}$$

Resubstitution rate

Misclassification counts at leaves:

- Left leaf error = 54 - 49 = 5
- Right leaf error = 46 - 45 = 1
- Total errors = 5 + 1 = 6

$$\text{Resubstitution rate} = 6 / 100 = 0.06 = 6\%$$

Minimum α for this split to be pruned

Using cost-complexity pruning: prune the split when the pruning decreases the cost:

$$\text{prune if } \alpha \geq \frac{R_{pruned} - R_{subtree}}{L - 1}$$

Where R is the impurity measured as a resubstitution error, and L is the number of leaves in the subtree (L = 2).

- R subtree = current subtree error = resubstitution = 0.06
- R pruned = error if parent kept as single leaf = $\frac{100-50}{100} = 0.5$

So $\alpha_{min} = \frac{0.5 - 0.06}{2 - 1} = 0.44$ therefore if $\alpha \geq 0.44$ the split will be pruned equivalently to keep the split you need $\alpha < 0.44$.

DATA ANALYSIS

The goal of this exercise is to predict whether a bank client will subscribe to a deposit product based on a telemarketing campaign. We have applied four supervised learning models:

1. Logistic Regression
2. Support vector classifier (SVC, linear kernel)
3. Random forest classifier
4. Gradient boosting classifier (used as a proxy for XGBoost)

The variable duration has been removed from the given dataset as it is not available prior to making the call and therefore can't be used for prediction. All categorical features were encoded using one-hot encoding, and numeric features were standardized when required (Logistic regression and svc). The dataset was split into training and testing sets, given the following results:

None

--- Models evaluation ---

Logistic Regression Test Error: 0.2989	// 29.89%
SVC (Linear Kernel) Test Error: 0.3049	// 30.49%
Random Forest Test Error: 0.2810	// 28.10%
Gradient Boosting (XGB Proxy) Test Error: 0.2687	// 26.87%

$$\text{Test Error} = 1 - \text{Accuracy}$$

Comments on the results

After obtaining the results, we may comment on them, explaining how they worked this way and key concepts about them:

Gradient Boosting

This model outperformed all the others. This result is consistent with theoretical explanations explained in lecture slides: boosting algorithms build trees sequentially, where each new one corrects previous tree errors. Several factors justify the its superior performance:

- Complexity of customer behavior: the variables interact in a nonlinear way.
- Boosting's ability to model heterogeneous effects: it learns fine-grained decision boundaries, making it particularly effective for tabular datasets with many categorical variables.
- Handling of imbalanced or noisy patterns: dataset may contain weak or noise signals, and this model is known to perform well in such environments by reweighting misclassified instances.

Random forest

It presented a strong performance but did not match Gradient boosting. Its relative success is due to its use of ensemble averaging over multiple decision trees, which:

- Handles categorical one-hot encoded variables naturally.
- Reduces overfitting through bootstrap aggregation.
- Captures nonlinear relationships and interactions between variables.

However, random forest does not benefit from sequential error correction, unlike boosted models. Each tree in a random forest is built independently, meaning the model is less capable of focusing on difficult samples. This explains the slightly lower predictive accuracy compared to the previous one.

Linear models

They both displayed the worst outcomes: highest test errors, which indicates that linear methods are poorly suited for this dataset. Several reasons justify this:

- Decision boundaries are insufficient: Customer behavior in marketing rarely follows a linear pattern, features interact in complex ways and linear classifiers can't capture such nonlinear interactions as easily as the other models
- Strong effect of categorical variables: the dataset contains many nominal variables and after one hot encoding, these generate high-dimensional sparse features. We notice that linear models struggle to assign meaningful weights when:
 - Categorical feature group interact
 - Effects vary across subgroups
 - Decision boundaries are piecewise or hierarchy based rather than linear based.
- Imbalance predictive patterns: marketing predictions tend to introduce many weak predictors rather than a few strong ones

In conclusion, our code demonstrates empirically what we have learned theoretically, tree-based ensemble models are better suited for predicting deposit subscriptions in bank telemarketing campaigns.

APPLICATION QUESTIONS

You are working for a lending start-up. Your business obtains financing from banks (variable rate based on 3 month EURIBOR in EUR) and provides short term (max for 6 month) point of sale financing to various e-commerce sites across the globe. For example, one of your largest clients is a US e-commerce site, which sells specialized tools for construction industry self-employed contractors.

a. Please describe the top risks your company is facing.

→ CREDIT RISK

This represents the possibility that borrowers fail to repay their short-term loans in full or on time. The risk is elevated because many clients like small contractors or international consumers may have limited credit histories, volatile income, or exposure to sector specific downturns.

→ LIQUIDITY RISK

The company may face difficulty meeting its short term obligations, particularly repayments to its financing banks. A sudden increase in overdue loans, delays in customer repayments or unexpected spikes in funding needs could create liquidity shortages.

→ INTEREST RATE RISK

The company's financing costs are linked to the 3-month EURIBOR. Any upward movement in this benchmark rate increases the cost of capital. Because the loans extended to customers often have fixed rates, a rising interest rate environment can significantly compress margins or even lead to negative profitability.

→ CURRENCY RISK

The company provides global finance to its clients, and the funding is done in Euros (EUR) but often repaid in foreign currencies. Because repayments to the company are converted from foreign currencies back to EUR, fluctuations in exchange rates can significantly change the EUR equivalent of a repayment. If EUR appreciates relative to USD, then the company's revenue from that client in EUR will be decreased and profits will suffer as a result.

→ COMPLIANCE RISK

The business is deeply reliant upon digital methods, cross-border integration of systems, and automated credit decisions. Failures of internal systems, fraudulent activity relating to borrower onboarding processes, errors made when processing data and non-compliance with regulatory requirements (such as KYC and AML) or data privacy legislation have the potential to result in financial loss and/or legal penalties being incurred.

b. How data analysis methods could help you assess and manage these risks.

Financial institutions face several risks that must be measured and managed to ensure stability and compliance. We could use the following analysis methods for each risk:

1. Credit risk.- it could use a logistic regression to identify high risk borrowers as it enables setting credit limits and risk-adjusted pricing
2. Liquidity risk.- We could use gradient boosting as it forecasts cash inflows and outflows under normal and stressed conditions and helps plan liquidity buffers and contingent funding.
3. Interest rate risk.- Using a random forest model stimulates portfolio sensitivity to rate changes. Supports scenario analysis and risk mitigation strategies.
4. Currency risk.- Implementing gradient boosting models complex effects of exchange rate movements on repayments and guides hedging and currency exposure management.
5. Compliance risk.- logistic regression ensures transparent, interpretable risk metrics for regulatory reporting and helps maintain adherence to credit and operational guidelines.

Applying data analysis techniques to generate actionable insights for all possible risks allows for predictive, scenario-based, and quantitative assessments. When using predictive models properly, an institution can predict likely losses and develop the best means to mitigate those losses whilst still remaining compliant with applicable regulations. Additionally, using predictive models within an agency supports portfolio resilience and enhances decision making.

c. During the latest review, you noted an increase in overdue (not repaid on time) loans. It was decided to perform a comprehensive analysis of credit losses. Please describe the steps of this analysis process and models that you would use at each step.

Our start-up has noticed an increase in overdue loans, so it is essential to analyse credit losses systematically. Using data-driven models will help us understand the risk and take actions. Here are the following steps to take:

1. Data collection and preparation

We gather loan data including borrower profiles, repayment history, loan mounts... and then clean the data and create features such as past defaults and client region to make it ready for analysis.

2. Exploratory data analysis

We visualize overdue loans, examine correlations, which allows us to identify patterns and highlight groups of borrowers that may pose higher risk.

3. Credit risk modeling

We predict the probability of default using logistic regression while gradient boosting can capture nonlinear borrower behaviors and we can estimate loss given default with gradient boosting to assess potential losses if default occurs.

4. Portfolio level analysis

We aggregate PD and LGD to estimate expected credit losses per client, region or currency and we use cohort analysis to track how different groups of loans evolve over time, which helps in understanding portfolio-level risk.

5. Stress testing and scenario analysis

We simulate scenarios such as rising EURIBOR or an economic slowdown and apply random forest to estimate how these events could impact the portfolio.

6. Reporting and decision making

We compared expected vs actual losses and based on the analysis, recommend actions while using logistic regression to ensure the risk metrics remain interpretable and suitable for regulatory reporting.

By combining data preparation, analysis, predictive modeling and scenario testing we can quantify credit risk and take targeted actions. This approach strengthens portfolio resilience and supports regulatory compliance.

d. Discuss advantages and disadvantages of logistic regression vs other classification models for the purpose of the above data analysis.

Advantages logistic regression	Disadvantages logistic regression
Simple, interpretable, outputs probabilities, suitable for regulatory reporting, works well on smaller dataset.	Assumes linear relationship, may underperform on complex / nonlinear data

Model	Advantages vs logistic regression	Disadvantages vs logistic regression
Random forest	Captures nonlinear relationships and feature interactions, handles large datasets, robust to noise, often higher predictive accuracy	Less interpretable, harder to explain to regulators, computationally heavier, may overfit if not tuned
Gradient boosting	Highly accurate, models complex patterns, effective for PD and LGD, handles nonlinear dependencies	Complex to implement and tune, less transparent, prone to overfitting on small datasets
SVC	Effective for high-dimensional data, clear separating hyperplane if classes are linearly separable	Less interpretable, does not naturally output probabilities, struggles with large datasets

e. You plan to implement the machine learning model to deal with credit scoring (deciding which clients to grant credit and which to decline). Please explain briefly how machine learning could work in this situation and describe the key considerations in setting up such a model.

Machine learning for credit scoring works by training predictive models on historical loan data, including borrower profiles, repayment history, loan amounts, client region, transaction patterns and macroeconomic indicators. The model estimates the probability of default (PD) for each applicant, generating a risk score that guides approval, loan limits, or risk-adjusted pricing. This enables automated, consistent, and data-driven credit decisions while reducing subjective judgment.

Key considerations include:

- Feature engineering and selection: identify the most predictive variables
- Class imbalance handling: defaults are rarer than on-time payments
- Model interpretability: ensures compliance and explainability
- Evaluation metrics: use precision and recall to measure performance
- Bias and fairness: avoid discriminatory outcomes and ensure fair lending
- Overfitting prevention: apply cross-validation, regularization or early stopping
- Threshold setting: align approval decision with business risk appetite
- Monitoring and recalibration: update the model as borrower behavior or market conditions change.

By addressing these factors, a credit scoring model improves lending decisions, reduces losses, and provides actionable insights for managing portfolio risk.